Junyoung Park

PhD Applicant

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RESEARCH INTERESTS

My research lies at the intersection of structural dynamics, IoT sensing, and applied machine learning, with a focus on continuous monitoring and load-carrying capacity assessment of civil infrastructure. I develop high-resolution IoT sensors for long-term acceleration and strain measurements, cloud-based data pipelines for large-scale data management, and learning-based models that translate raw measurements into actionable indicators such as natural frequencies, displacement via data fusion, and vehicle load events. Beyond bridges, this approach extends to prefabricated structures during transportation and to port infrastructures, with the broader goal of enabling data-driven, proactive maintenance of safety-critical structures.

EDUCATION

Chung-Ang University, Seoul, Republic of Korea — Mar 2022 – Feb 2024

M.S. in Civil Engineering (Structural Engineering)

- Thesis: A Scalable Bridge Health Monitoring System using an IoT Sensor and Cloud Computing
- Advisor: the late Prof. Jong-Woong Park · GPA 4.39 / 4.5

Chung-Ang University, Seoul, Republic of Korea — Mar 2017 – Feb 2022

B.E. in Civil Engineering & B.S. in Mathematics (Double Degree)

- Thesis: Development of Low-power IoT Sensor and a Cloud-based Data Fusion Displacement Estimation Method for Ambient Bridge Monitoring
- GPA 4.20 / 4.5

RESEARCH EXPERIENCE

UDNS Co., Ltd., R&D Division, Yongin, Republic of Korea — Apr 2024 – Present

Researcher (Industrial Technical Personnel Program)

HS-WIM (Strip-Type High-Speed Weigh-in-Motion)

- Advanced a commercial product line already deployed in the market by designing signal-acquisition PCBs and developing YOLO-based load-segment detection coupled with FIR-filtered dynamic weight estimation for real-time, in-motion truck weighing. [Park et al. 2026, KSCE; Patent KR 10-2025-0068214]
- Conducted controlled indoor and on-road noise tests to validate robustness of the strip-WIM signal-processing pipeline against vehicle speed, axle configuration, and pavement conditions.

OBM (On-Board Mass) International Certification

- Led the Australian TCA (Transport Certification Australia) Category C type-approval campaign for the commercial OBM device: requirement analysis, technical documentation, environmental/EMC/IP-rated test design, accredited test management, and final certification reporting — enabling overseas market entry.

R&D Programs & Proposals

- Served as lead researcher on multiple Ministry of Land, Infrastructure and Transport (MOLIT) R&D projects (interim/final reports, cross-institutional coordination), delivering 2 journal papers, 3 patent applications, 3 software registrations, and 4 accredited test certificates.
- Authored public R&D proposals (MOLIT, Ministry of the Interior and Safety, related agencies); 3 of 4 submitted proposals were selected for funding.
- Led the renewal-review campaign for the Korean Excellent Logistics Technology designation (No. 2): application documents, technical presentation, on-site review.

Smart Infrastructure Technology Laboratory, Chung-Ang University — Jun 2019 – Feb 2024

Graduate Researcher (Advisor: the late Prof. Jong-Woong Park)

Long-Term Bridge Monitoring via IoT Sensors and Cloud Analytics

- Designed JANET, a 24-bit, 3-axis accelerometer + 8-channel strain IoT sensor with an event-driven power-management circuit (~1 mA standby) enabling continuous, multi-month monitoring on a single battery and direct LTE Cat-1 uplink to a cloud server without a gateway. [Park et al. 2026, Measurement (under review)]
- Deployed JANET on four operational bridges in Seoul (Seongsan Bridge, Paldang Overpass, Seosomun Overpass, Jamsu Bridge), with continuous data collection in service traffic since July 2023.
- Built an InfluxDB + MySQL data-management backbone with Grafana dashboards for non-expert bridge operators, and developed analysis pipelines for natural-frequency tracking, displacement estimation via sensor fusion, and substructure condition assessment. [Park et al. 2021, JCSEIK]

Precast-Structure Lifecycle Monitoring (Fabrication → Transport → Site)

- Developed an ultra-low-power, multi-channel strain visualization module with embedded E-paper QR display for whole-lifecycle quality assurance of precast members. [Khan, …, Park et al. 2021, Sensors; Patents KR 10-2022-0159449, KR 10-2024-0017869]
- Implemented a smartphone-based QR data acquisition pipeline via KakaoTalk, integrated with automated analysis reporting through the ChatGPT API for on-site operators.
- Developed a high-resolution IoT sensor system with a BLE-to-LTE gateway for 1:N transport monitoring, capturing acceleration, strain, and tilt during lifting/transport/loading. [Khayam, …, Park et al. 2023, Automation in Construction]

Smart Maintenance for Port Structures

- Developed IoT sensors and an AI-based event classifier for berthing/unberthing impact detection and long-term quay-wall behavior monitoring; built an MQTT real-time data pipeline with InfluxDB. [Shin, Park et al. 2024, JMSE]

Earlier Projects — Smart Earthquake Detection · Self-Sensing UHPC

- Designed a low-cost IoT seismic sensor with a two-stage STA/LTA + frequency-based detection algorithm to minimize false positives. [Won, Park et al. 2020, Sensors; Patent KR 10-2020-0059559]
- Co-developed a smart Ultra-High-Performance Concrete (UHPC) circuit for self-stress sensing, with a 1:N BLE beacon-based data-collection pipeline. [Le, …, Park et al. 2021, Sensors]

JOURNAL PUBLICATIONS

1. **Park, J.**, Shin, J., & Won, J. Development of a Portable Multimetric Sensing System for Bridge Load Testing and Load-Carrying Capacity Evaluation. *Measurement* (under review), 2026.
2. **Park, J.**, Kim, J., Cho, Y., & Jung, Y. (2026). YOLO-Based Automatic Load Segment Detection for High-Speed Weigh-in-Motion Systems. *Journal of the Korean Society of Civil Engineers*, *46*(1), 95–104.
3. Shin, J., **Park, J.**, Won, J., Park, J. W., & Min, J. (2024). Development of AI-Based Multisensory System for Monitoring Quay Wall Events. *Journal of Marine Science and Engineering*, *12*(11), 1902.
4. **Park, J.**, Kim, J., Kim, J., & Cho, Y. (2024). Stabilization of Dynamic Weight Estimation Systems for Commercial Vehicles during Driving. *Journal of the Korean Society of Mechanical Technology*, *26*(5), 1041–1046.
5. Khayam, S. U., Ajmal, A., **Park, J.**, Kim, I. H., & Park, J. W. (2023). Tendon Stress Estimation from Strain Data of a Bridge Girder Using Machine Learning-Based Surrogate Model. *Sensors*, *23*(11), 5040.
6. Khayam, S. U., Won, J., Shin, J., **Park, J.**, & Park, J. W. (2023). Monitoring Precast Structures during Transportation Using a Portable Sensing System. *Automation in Construction*, *145*, 104639.
7. Won, J., Park, J. W., **Park, J.**, Shin, J., & Park, M. (2021). Development of a Reference-Free Indirect Bridge Displacement Sensing System. *Sensors*, *21*(16), 5647.
8. Le, H. V., Kim, T. U., Khan, S., **Park, J. Y.**, Park, J. W., Kim, S. E., … & Kim, D. J. (2021). Development of Low-Cost Wireless Sensing System for Smart Ultra-High Performance Concrete. *Sensors*, *21*(19), 6386.
9. Khan, S., Won, J., Shin, J., **Park, J.**, Park, J. W., Kim, S. E., … & Kim, D. J. (2021). SSVM: An Ultra-Low-Power Strain Sensing and Visualization Module for Long-Term Structural Health Monitoring. *Sensors*, *21*(6), 2211.

10. **Park, J.**, Shin, J., Won, J., Park, J. W., & Park, M. (2021). Development of Low-Power IoT Sensor and Cloud-Based Data Fusion Displacement Estimation Method for Ambient Bridge Monitoring. *Journal of the Computational Structural Engineering Institute of Korea*, *34*(5).
11. Won, J., **Park, J.**, Park, J. W., & Kim, I. H. (2020). BLESeis: Low-Cost IoT Sensor for Smart Earthquake Detection and Notification. *Sensors*, *20*(10), 2963.

CONFERENCE PROCEEDINGS

1. **Park, J.**, Shin, J., & Park, J. W. (2023). Cloud-Based Ambient Bridge Monitoring using Strain and Acceleration Measurements. *Proc. ITC-CSCC*.
2. **Park, J.**, Shin, J., Jang, H., & Park, J. W. (2022). Development of IoT Sensor System and Web-Based Document Automation for Bridge Load-Bearing Capacity Evaluation. *Proc. CONVR*.

PATENTS

1. Park, J. W., **Park, J.**, & Shin, J. *Continuous Bridge Load-Carrying Capacity Assessment System and Method*. KR 10-2025-0174354. Application filed Nov 18, 2025.
2. Park, J. W., **Park, J.**, & Shin, J. *Automatic Load-Event Interval Recognition Using IoT Sensors for Structural Capacity Assessment*. KR 10-2853678. Registered Aug 28, 2025.
3. Park, J. W., **Park, J.**, & Kim, J. *Real-Time Dynamic Truck Operating Weight Correction Based on an FIR Filter*. KR 10-2025-0068214. Application filed Dec 30, 2024.
4. Park, J. W., **Park, J.**, & Shin, J. *Concrete Crack Trajectory Monitoring and Long-Term History Visualization*. KR 10-2024-0017869. Registered Feb 6, 2024.
5. Park, J. W., **Park, J.**, & Shin, J. *IoT Sensor-Based Integrated Monitoring for Precast Transportation*. KR 10-2022-0159449. Registered Nov 24, 2022.
6. Park, J. W., Won, J., **Park, J.**, & Shin, J. *Cloud-Based IoT Sensor Data Preprocessing and Deep-Learning Anomaly Detection*. KR 10-2022-0021506. Registered Feb 18, 2022.
7. Park, S., Shin, J., & **Park, J.** *Citizen-Participatory Smart City System Using Sensor Measurement and Data Visualization*. KR 10-2021-0126695. Registered Sep 24, 2021.
8. Won, J., **Park, J.**, & Park, J. W. *Smart IoT Earthquake Sensor Unit and Notification Method*. KR 10-2020-0059559. Registered May 19, 2020.

HONORS & AWARDS

Best Paper Presentation Award, Korea Institute for Structural Maintenance and Inspection	Apr 2024
Gold Award, X-Corps Plus Festival, National Research Foundation of Korea	Nov 2022
CAU Graduate Research Scholarship, full tuition, Chung-Ang University	Mar 2022
SeAH Haiam Scholarship, full undergraduate tuition (two years)	Jun 2020
Minister's Award, Smart Earthquake Detection Competition, Ministry of the Interior	Feb 2019
Grand Prize, AI Object Recognition Hackathon, 2018 Engineering Festival	Nov 2018

SKILLS

Programming: Python, MATLAB, C, JavaScript/Node.js, SQL
ML: PyTorch, scikit-learn, pandas, NumPy, YOLO
Software: MATLAB/Simulink, SolidWorks, Altium Designer
Data: AWS, Docker, MySQL, InfluxDB, LabVIEW
Languages: English (TOEFL iBT 100/120, 2026), Korean (native)